PEN & PAPER

June 8th, 2026

- PROBLEM SOLVING -

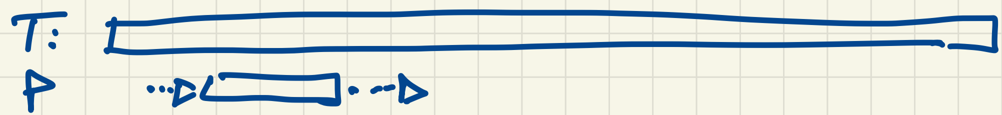
PATTERN MATCHING

THE PROBLEM

<u>Input</u>: text T of length n, pattern P of length m

<u>Output</u>: occurrences of P in T

"BRUTE-FORCE" METHOD



$O(n \cdot m)$ COMPARISONS : can we do better ?!

.... C A G A G A G T A G A ....

Text

Pattern AGAGAT

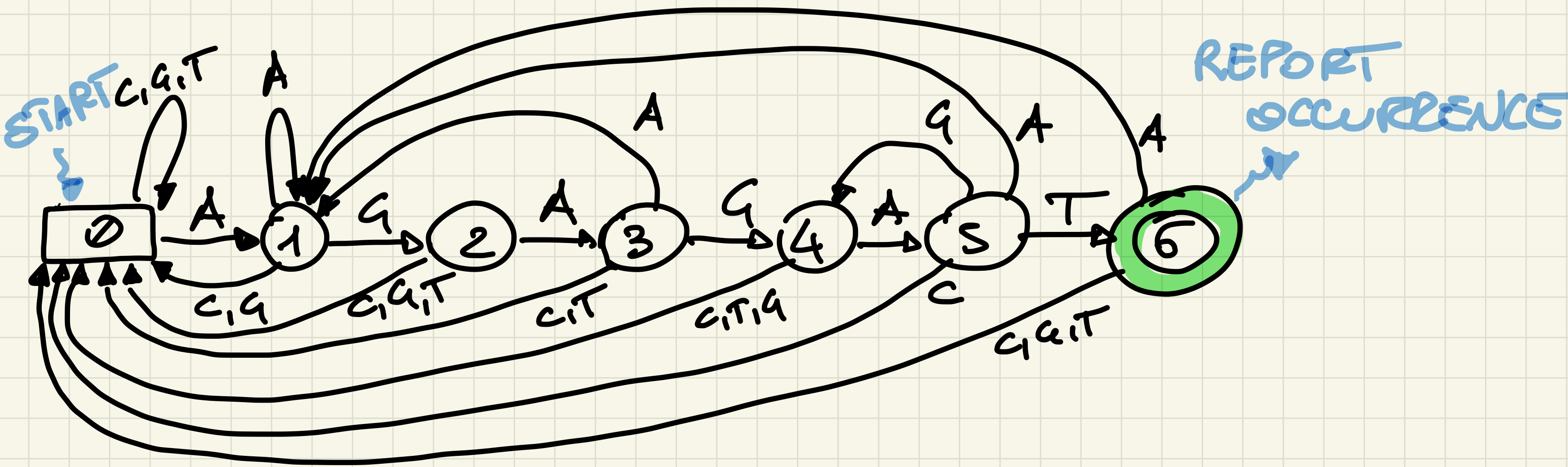
A

→ A G A G A T ..

→ A G A G A

→ A G A ....

AUTOMATA



IT'S YOUR TURN!

$P_1 = CAT$

$P_2 = CTG$

BUILD THE AUTOMATA!

$$P = A^* \left\{ \begin{matrix} C \\ G \end{matrix} \right\} T \left\{ \begin{matrix} A \\ CA \end{matrix} \right\} C$$

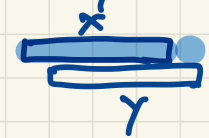
BUILD THE AUTOMATA!

DE BRUIJN GRAPHS

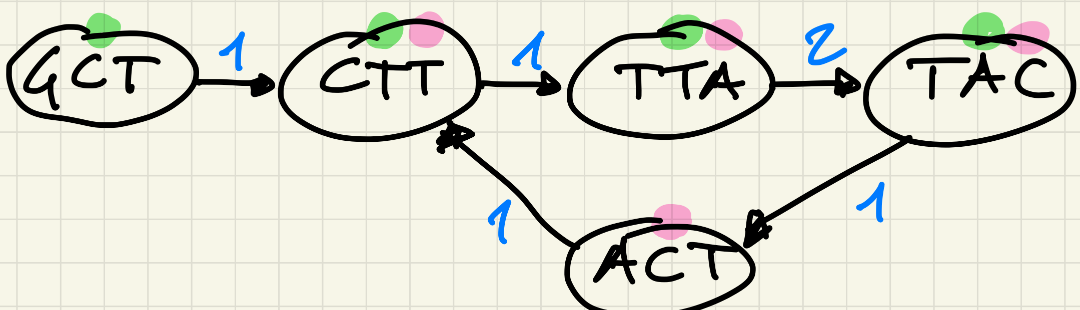
$\boxed{k\text{-dBG}}$

$\Sigma = \{A, C, G, T\}$

given k
and given a set S of sequences

- A node $\forall$ k-mer $\in S$
- Edge $(x) \rightarrow (y)$ iff x and y overlap by $k-1$ letters and  $\in S$

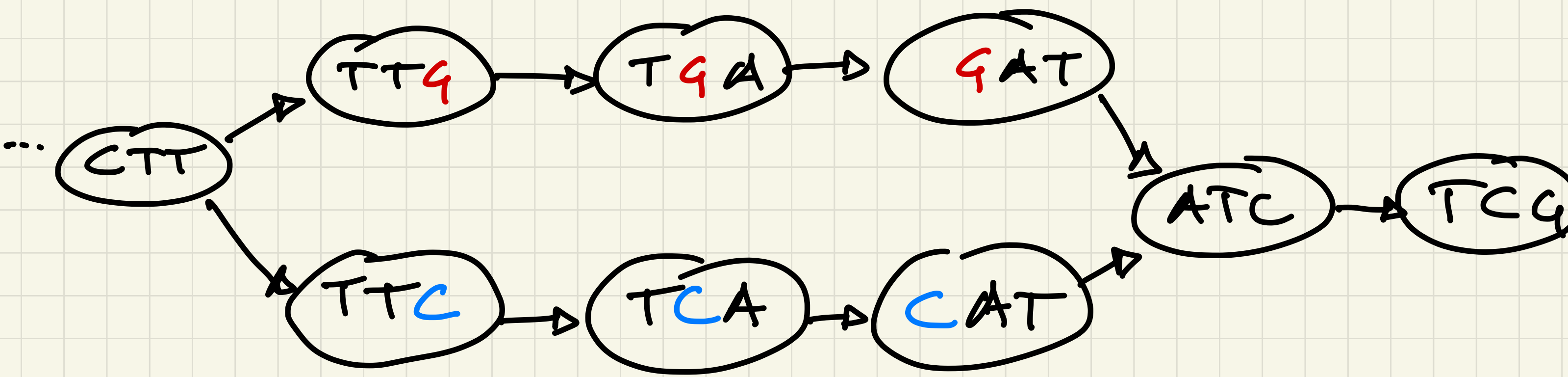
$S = \{GCTTAC, TTACTT\}$



SNPs in dBG :

S : ... CTTGATCG... (s_1)

... CTTCATCG... (s_2)



TWO ALTERNATIVE PATHS OF k NODES FROM A COMMON SOURCE TO A COMMON TARGET

IT'S YOUR TURN !

WHAT IF I HAVE AN INDEL ?

S: $\cdots$ TTC $\overbrace{\text{TTGAA}}^{\text{length } \ell}$ CGT.. (S_1)

$\cdots$ TTCCGT (S_2)

$x = 3$

WHAT IF I HAVE A FRAGMENT SUBSTITUTION ?
(e.g. AS in mRNA)

S_1 $\cdots$ CGT F_1 TCG $\cdots$

S_2 $\cdots$ CGT F_2 TCG $\cdots$

$|F_1| = \ell_1$ $\quad$ $|F_2| = \ell_2$

EFFICIENT ALGORITHMS ON ARRAYS

"PSEUDOCODE" OF BINARY SEARCH (iterative version)

```
BS(A, p, r, k) // BS(A, 0, n-1, k)
IF ((k < A[p]) || (k > A[p]))
    THEN RETURN -1;
WHILE (p<r) DO
    { q = (p+r)/2 ;
      IF (A[q]==k) THEN RETURN q;
      IF (A[q]>k) THEN r = q-1
                  ELSE p = q+1
    }
RETURN -1
```

Input: SORTED A ; VALUE K

Output: $i : A[i] = K$

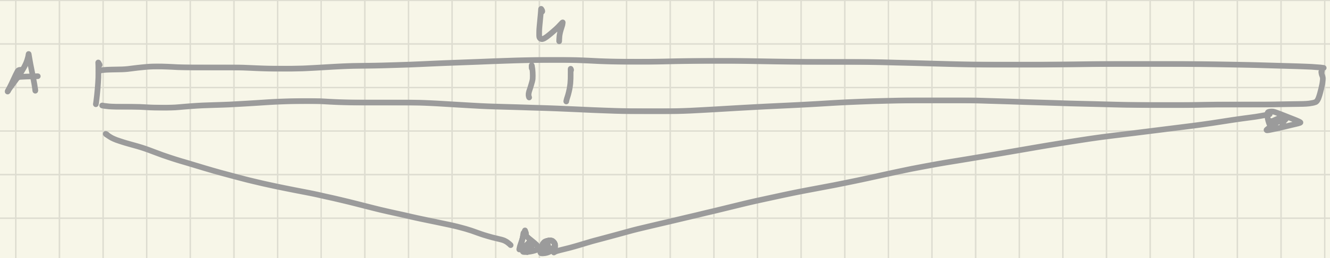
-1 if $K \notin A$

IT'S YOUR TURN !

Input : A array : $\exists\ 0 \le k \le n-1$ such that

$$A[0] > \dots > A[k] < A[k+1] < \dots < A[n-1]$$

Output : k



WRITE AN EFFICIENT PSEUDOCODE !

12 COINS

YOU HAVE 12 APPARENTLY IDENTICAL COINS
BUT AT MOST ONE OF THEM IS ACTUALLY "SPECIAL"
AND HENCE LIGHTER OR HEAVIER

YOU HAVE A TWO PLATES SCALE
(=> 3 POSSIBLE OUTCOMES

/ — \ )

DESIGN A PROCEDURE TO FIND THE SPECIAL COIN (IF THERE) USING THE SCALE AT MOST 3 TIMES

SEQUENCE ALIGNMENT

Input: X, Y sequences

Output: the best alignment of X and Y

E.G. MINIMUM EDIT DISTANCE

OR

MAXIMUM SIMILARITY SCORE

THERE IS AN EXPONENTIAL NUMBER OF POSSIBLE ALIGNMENTS!

CHECK ANTOINE'S CLASS TOMORROW!